

Digital Employee Platform

System Design Diagrams — Reference Document

This document contains all architectural diagrams for the Digital Employee platform: a multi-tenant AI-powered customer service system built with .NET 9 / ASP.NET Core 9 Clean Architecture.

Section 1: Entity–Relationship Diagrams (Modern Style) — *6 domain groups + full overview*

Section 2: Traditional Chen-Notation ERDs — *5 domain groups + full combined*

Section 3: UML Class Diagrams — *6 domain groups + combined overview*

Section 4: UML Use Case Diagram — *System actors and use cases*

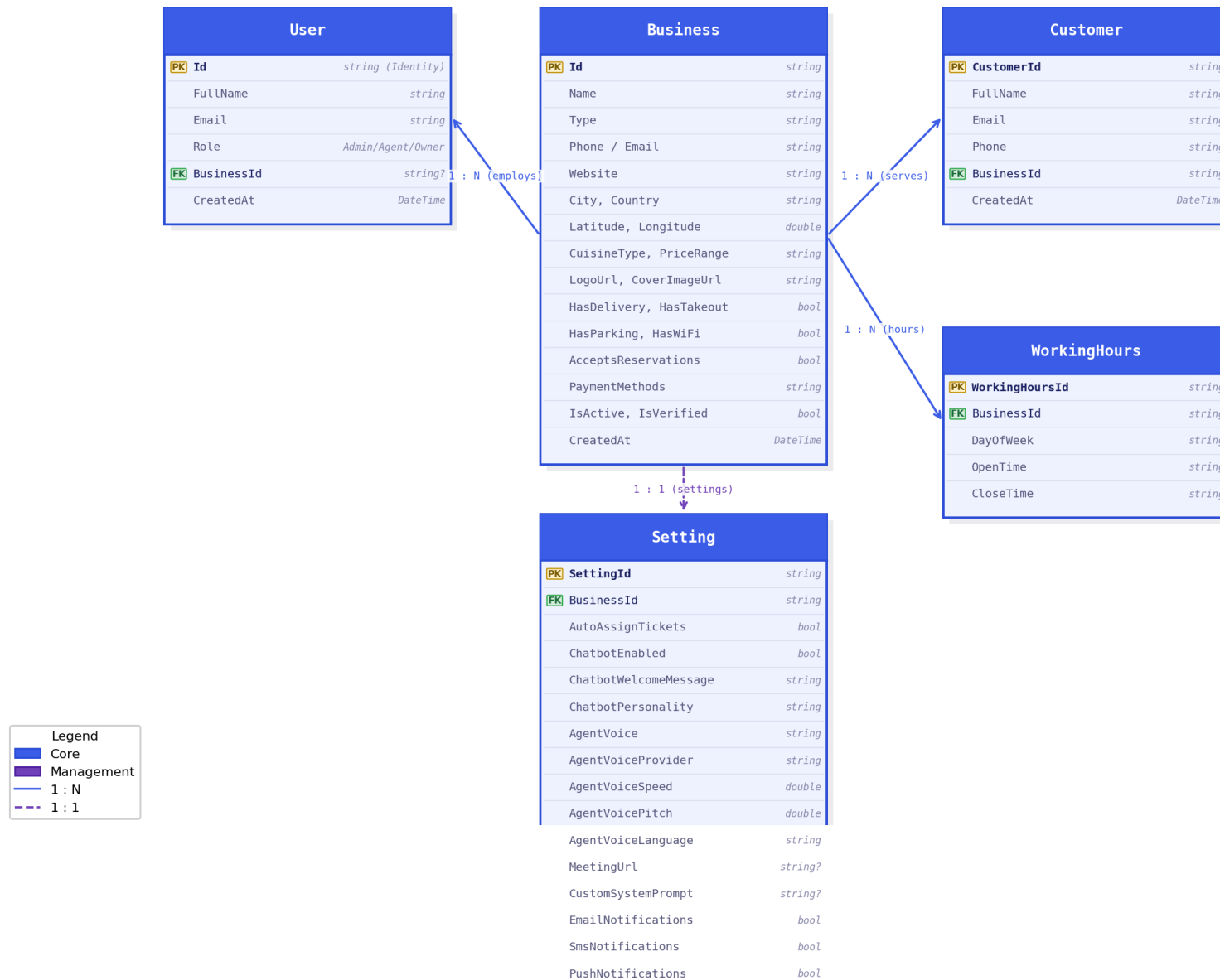
Section 1 — Entity–Relationship Diagrams (Modern)

Modern-style ERDs using coloured boxes. Each entity shows its table name, primary key (■), foreign keys (FK), and key attributes. Six domain groups plus a full-system overview.

Core Entities — Business, User, Customer, WorkingHours, Setting

ERD — Group 1: Core Entities

Business is the central multi-tenant entity. Every other entity belongs to a Business.

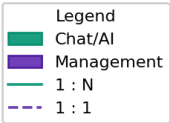
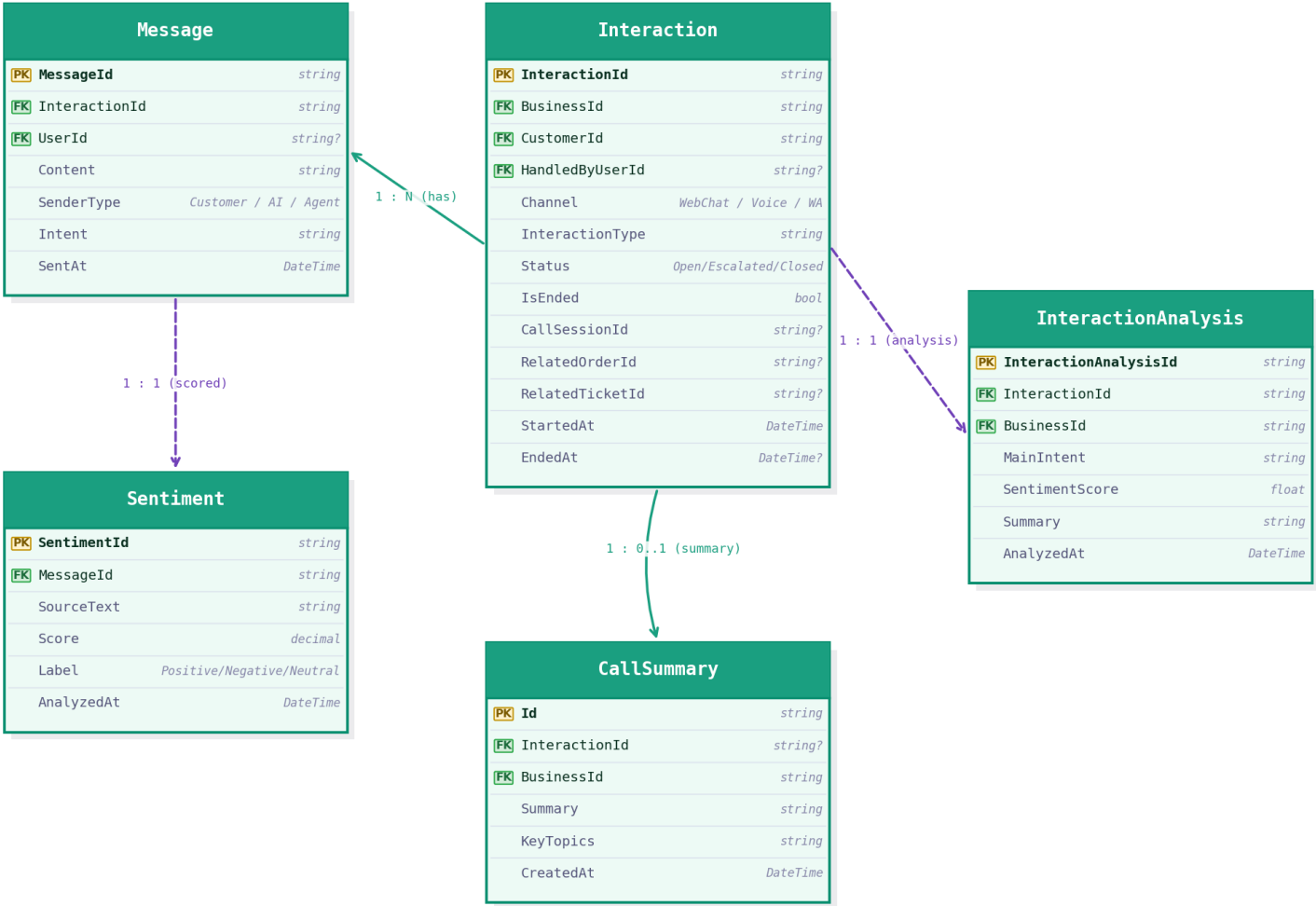


Core Entities — Business, User, Customer, WorkingHours, Setting

Chat & AI — Interaction, Message, Sentiment, Analysis, CallSummary

ERD — Group 2: Chat & AI Interaction Entities

Tracks every customer conversation (chat & voice), messages, AI analysis and call summaries.

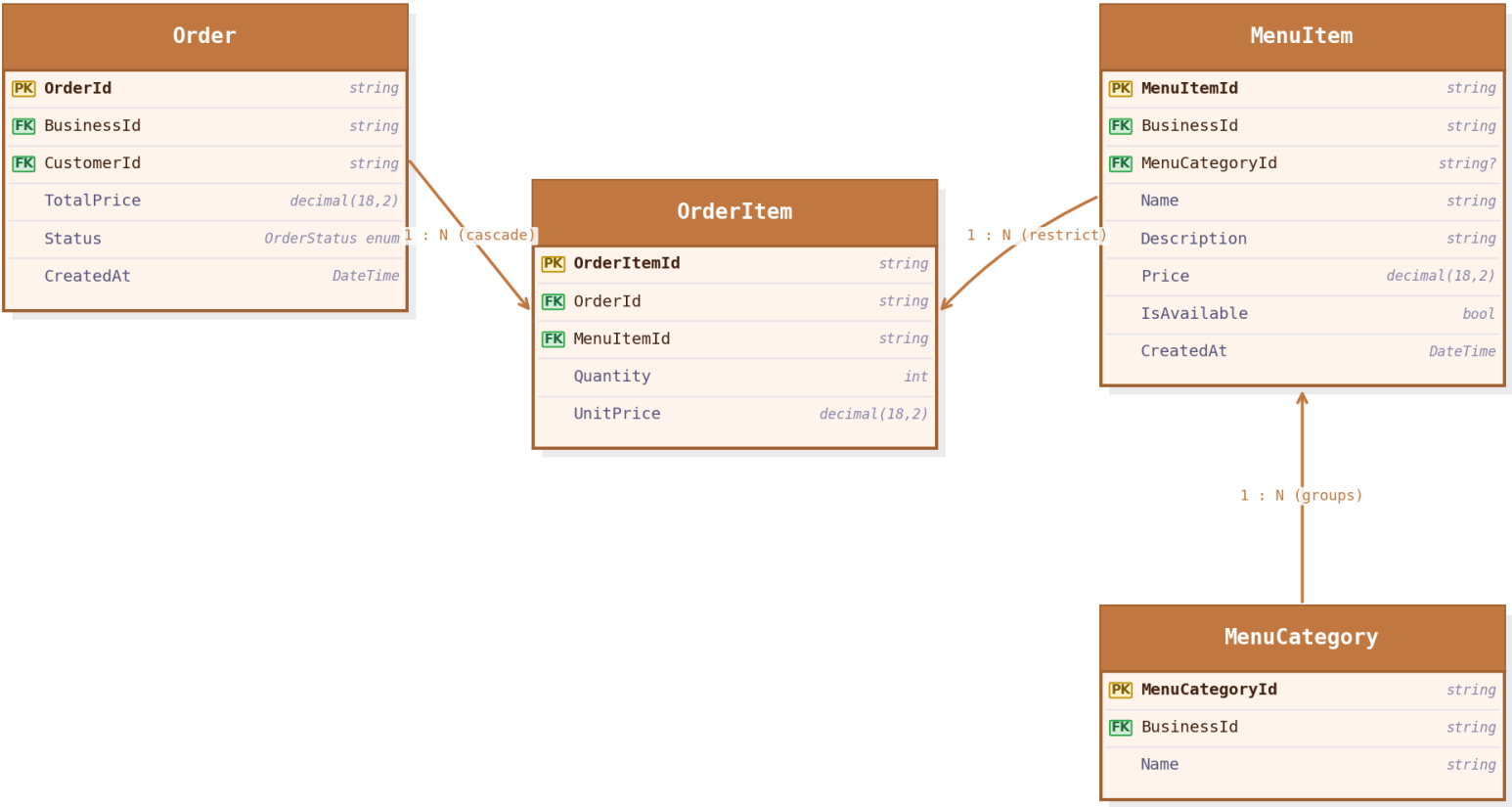


Chat & AI — Interaction, Message, Sentiment, Analysis, CallSummary

Orders & Menu — Order, OrderItem, MenuItem, MenuCategory

ERD — Group 3: Orders & Menu Entities

Manages customer orders, order items, menu catalogue and categories.



Legend

Orders

1 : N

Orders & Menu — Order, OrderItem, MenuItem, MenuCategory

Support — Ticket, Feedback, Notification

ERD — Group 4: Support Entities

Handles support tickets raised by customers, feedback collection, and user notifications.

Notification	
PK NotificationId	string
FK BusinessId	string
FK UserId	string?
Title	string
Message	string
IsRead	bool
CreatedAt	DateTime

Ticket	
PK Id	string
FK BusinessId	string
FK CustomerId	string
FK AssignedToUserId	string?
Subject	string
Status	Open/InProgress/Closed
TicketType	string
PriorityLevel	Low/Normal/High/Critical
EscalationConfidence	double 0-1
EscalationReason	string?
ResolutionNotes	string?
InteractionId	string?
RelatedOrderId	string?
IsEnded	bool
CreatedAt	DateTime
ClosedAt	DateTime?

Feedback	
PK FeedbackId	string
FK CustomerId	string
FK TicketId	string?
FK InteractionId	string?
Comment	string
Rating	int 1-5
SentimentScore	double
CreatedAt	DateTime

1 : N (receives)

Legend

Support

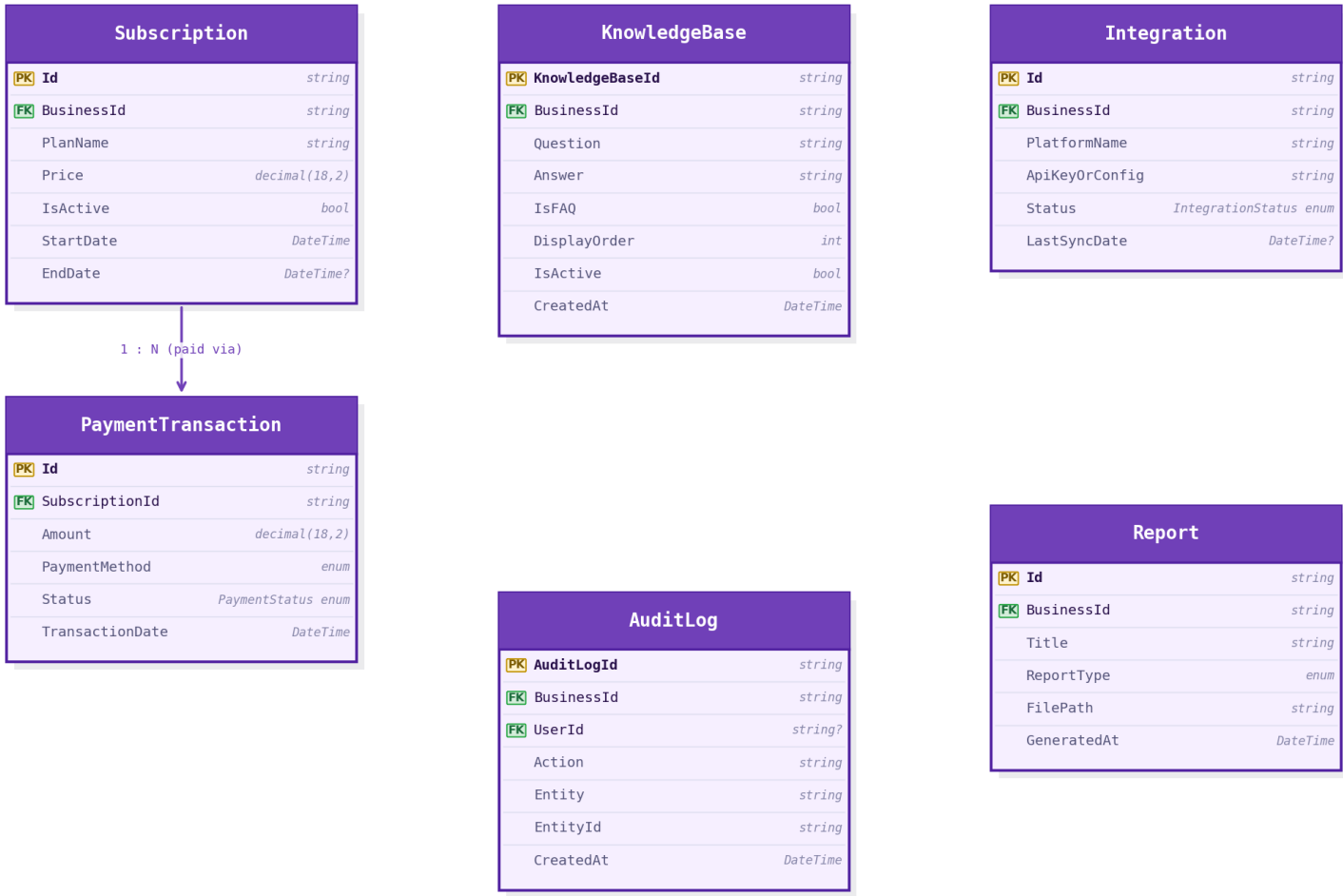
1 : N

Support — Ticket, Feedback, Notification

Management — Subscription, Payment, KnowledgeBase, Integration, AuditLog, Report

ERD — Group 5: Business Management Entities

Handles subscriptions, payments, knowledge base, integrations, audit logs, and reports.

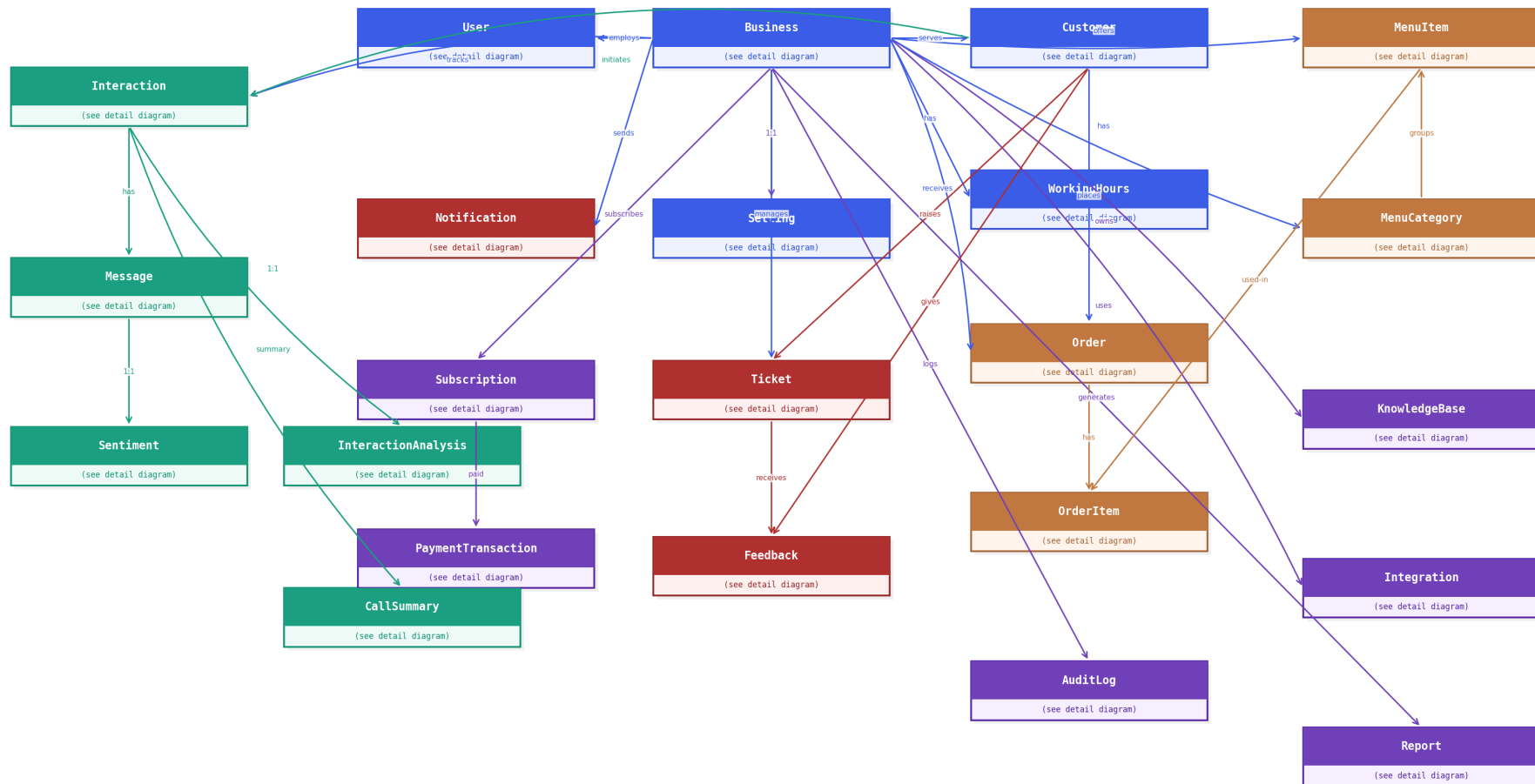


Management — Subscription, Payment, KnowledgeBase, Integration, AuditLog, Report

Full System Overview — all 23 entities

ERD — Full Overview: All 23 Entities & Relationships

Business (center-top) is the multi-tenant anchor. Lines show FK dependencies.



Full System Overview — all 23 entities

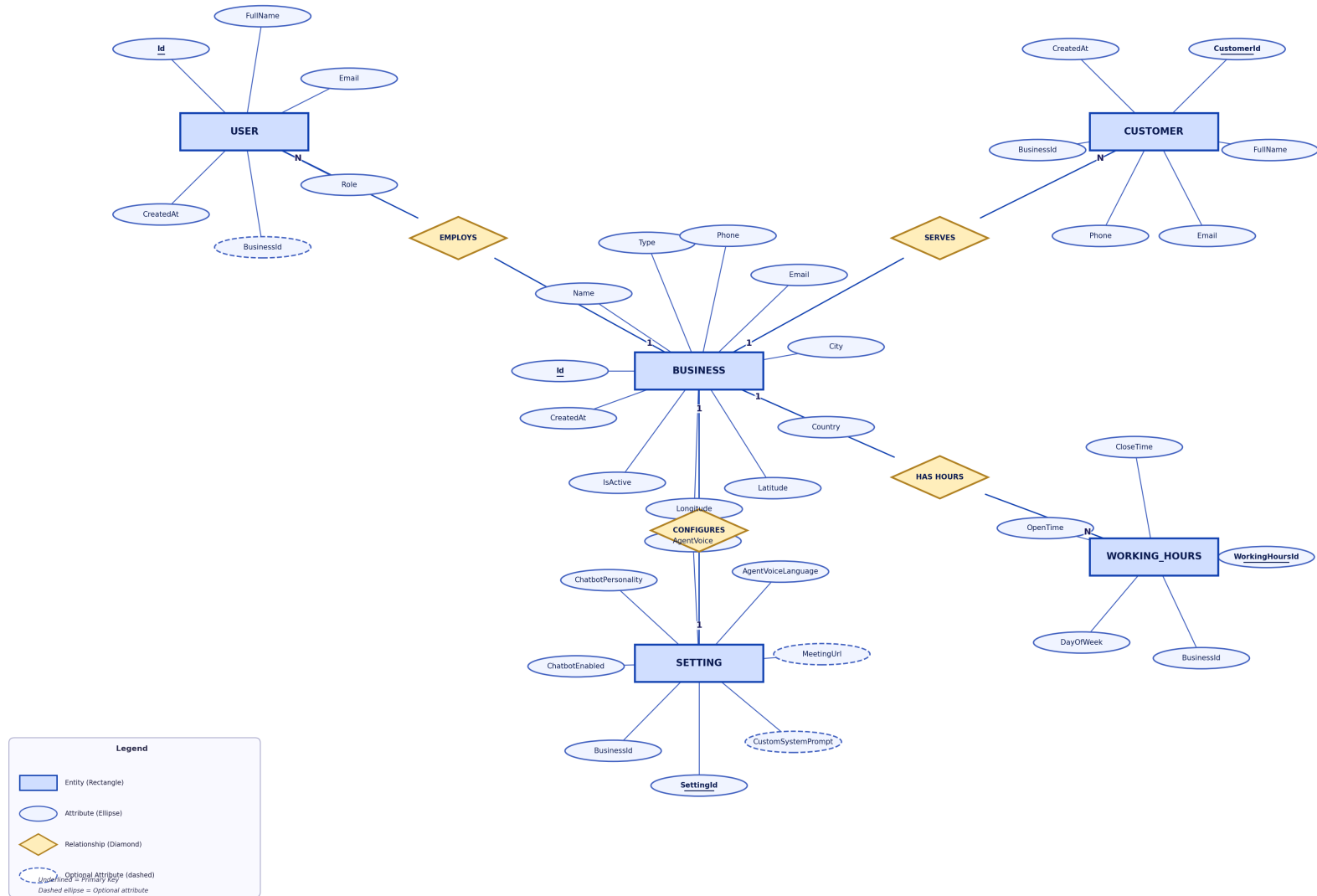
Section 2 — Traditional Chen-Notation ERDs

Traditional Chen notation: rectangles = entities, ellipses = attributes (underlined = PK, dashed = optional/nullable), diamonds = relationships, 1/N cardinality labels on each line.

Chen ERD — Core Domain

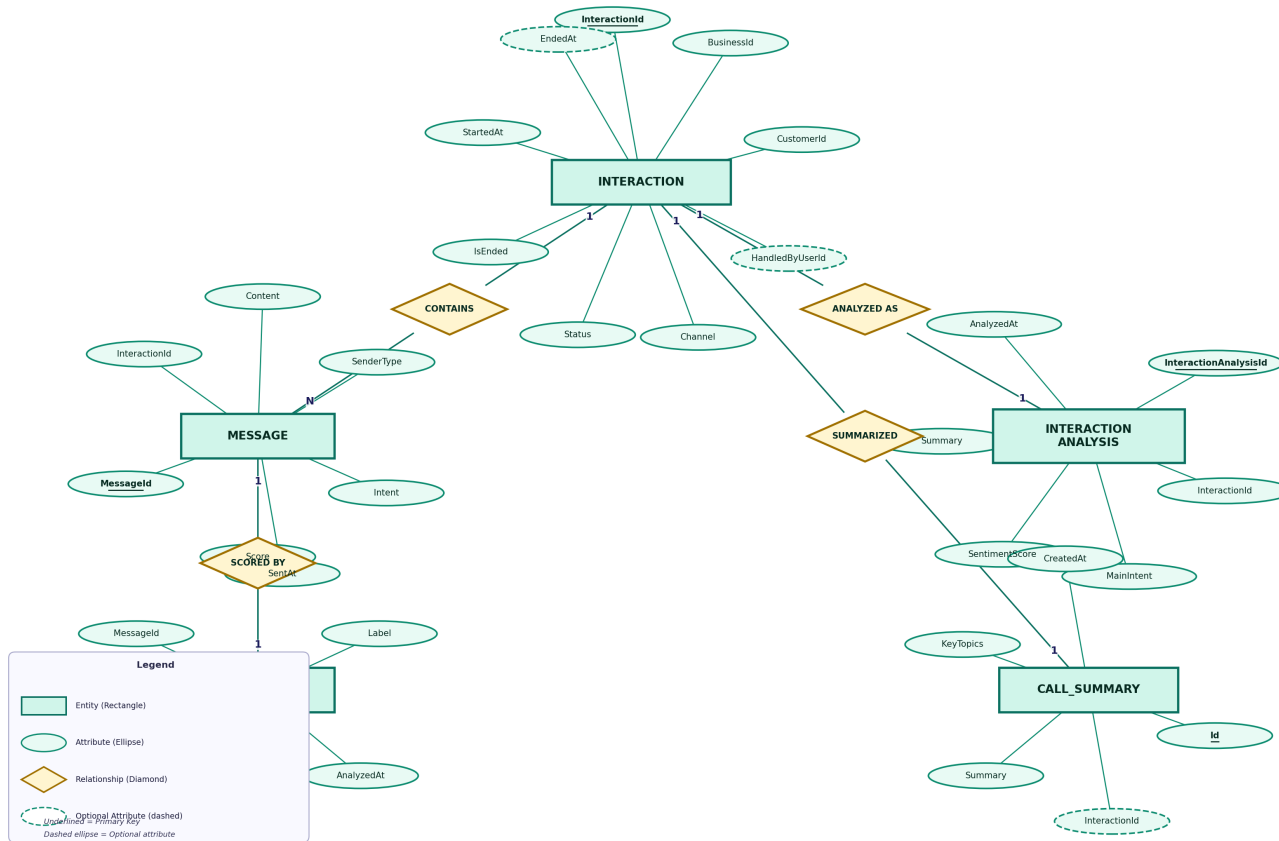
ERD — Group 1: Core Entities (Chen Notation)

Rectangles = Entities | Ellipses = Attributes | Diamonds = Relationships | Underline = PK



Chen ERD — Core Domain

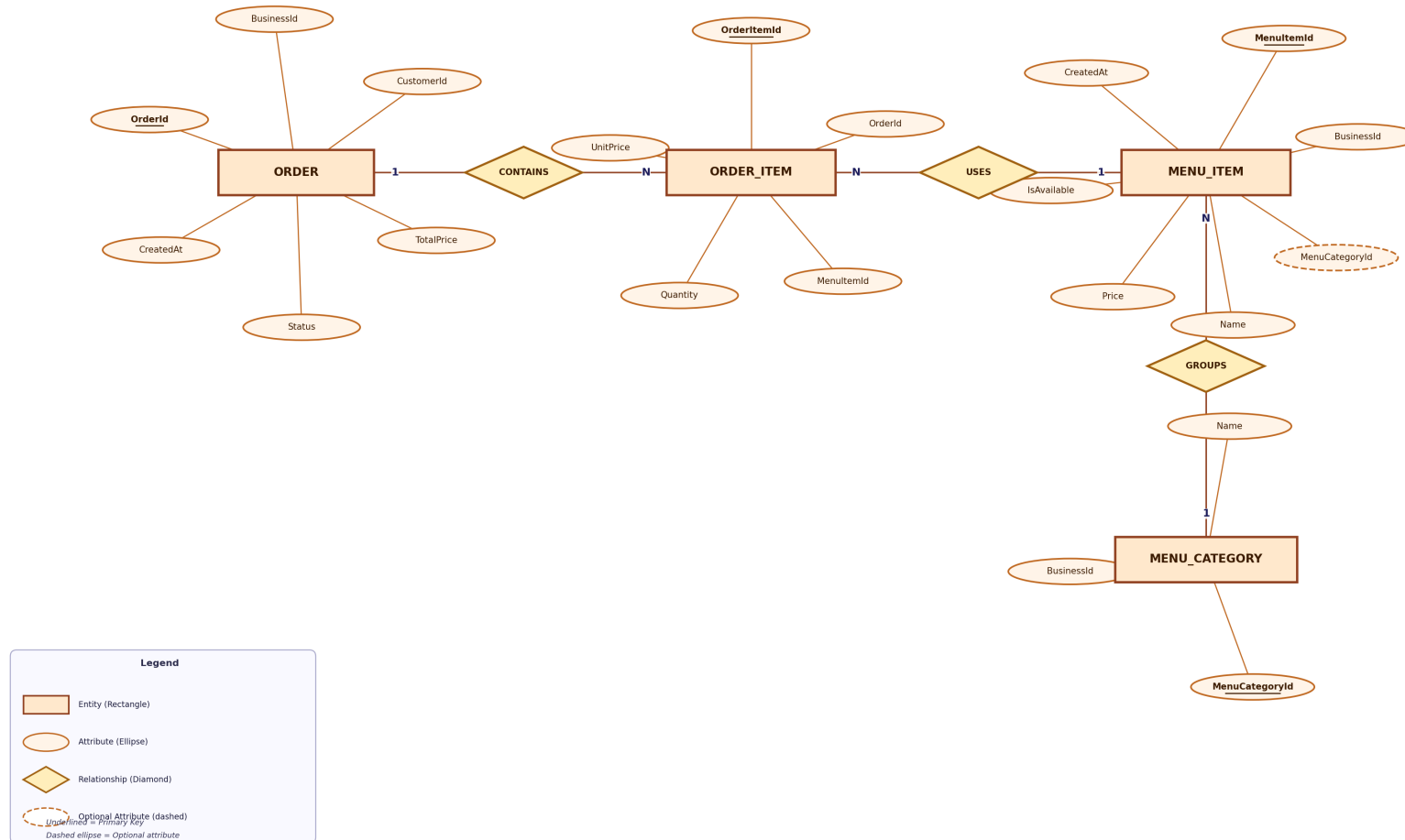
ERD — Group 2: Chat & AI Interaction (Chen Notation)



Chen ERD — Chat & AI Domain

Chen ERD — Orders & Menu Domain

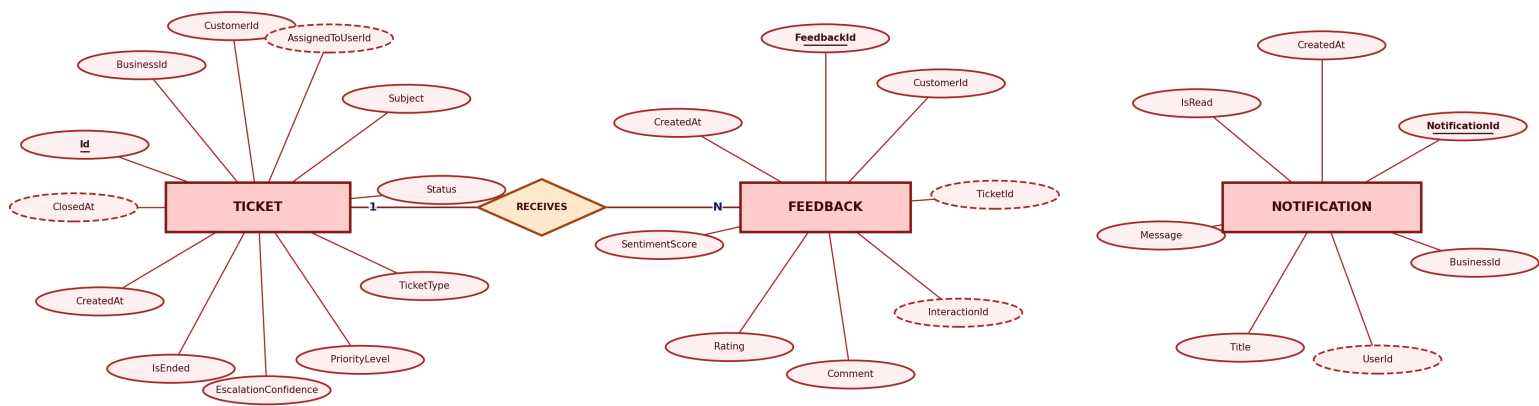
ERD — Group 3: Orders & Menu (Chen Notation)



Chen ERD — Orders & Menu Domain

Chen ERD — Support & Notifications Domain

ERD — Group 4: Support (Chen Notation)



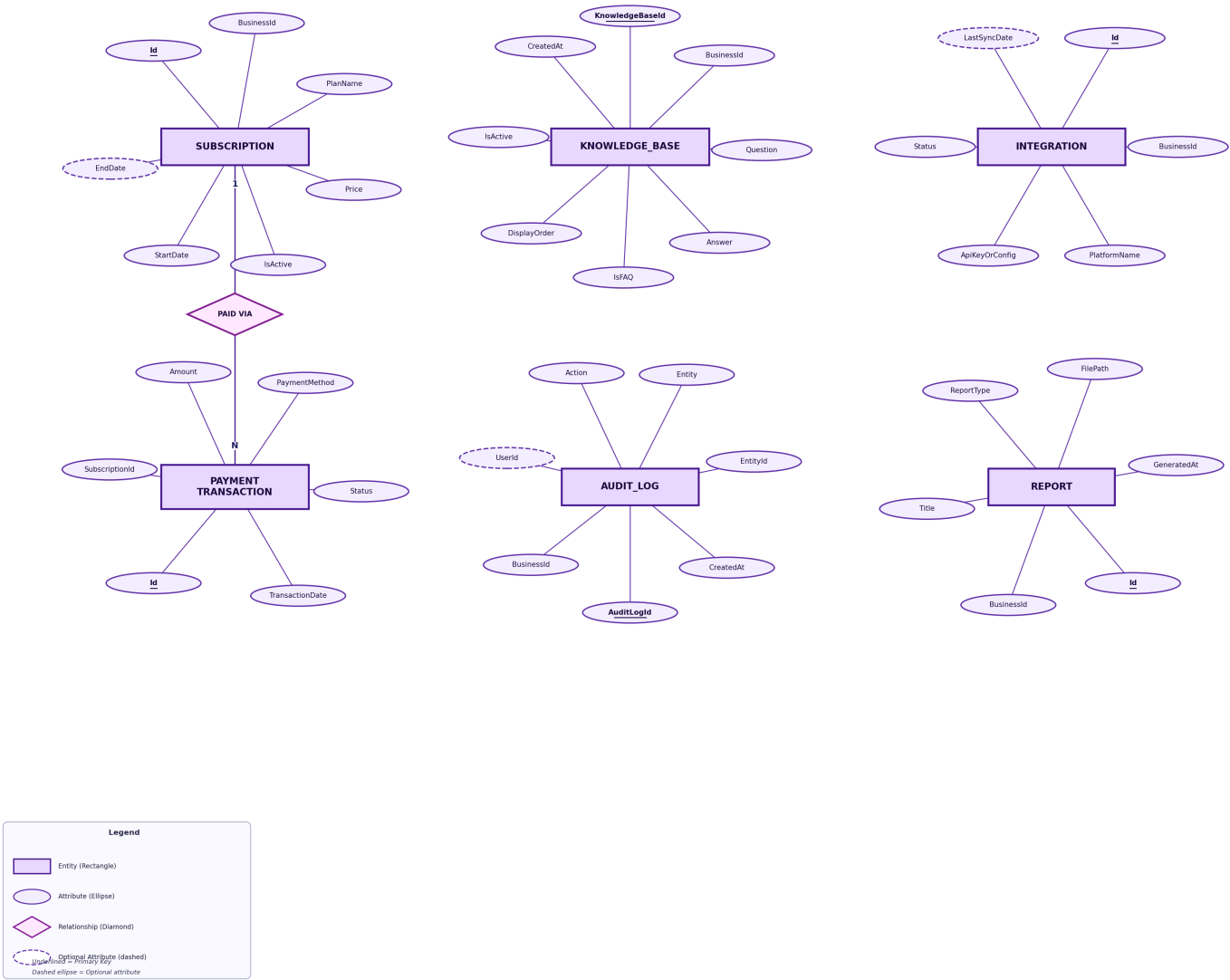
Legend

- Entity (Rectangle)
- Attribute (Ellipse)
- Relationship (Diamond)
- Optional Attribute (dashed)
- Underlined = Primary Key
- Dashed ellipse = Optional attribute

Chen ERD — Support & Notifications Domain

Chen ERD — Business Management Domain

ERD — Group 5: Business Management (Chen Notation)

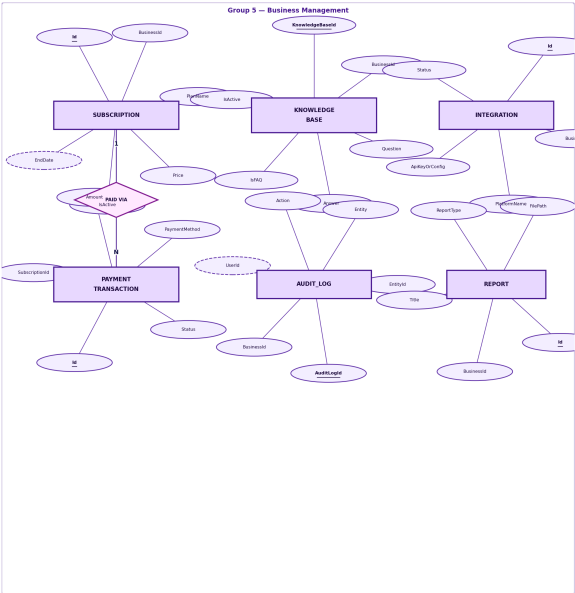
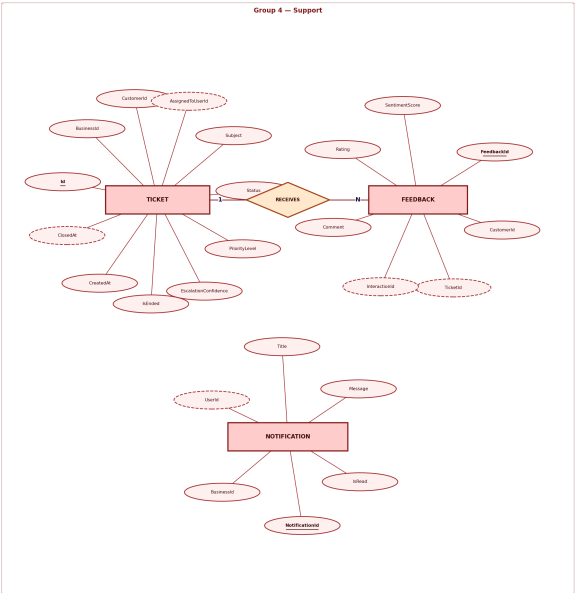
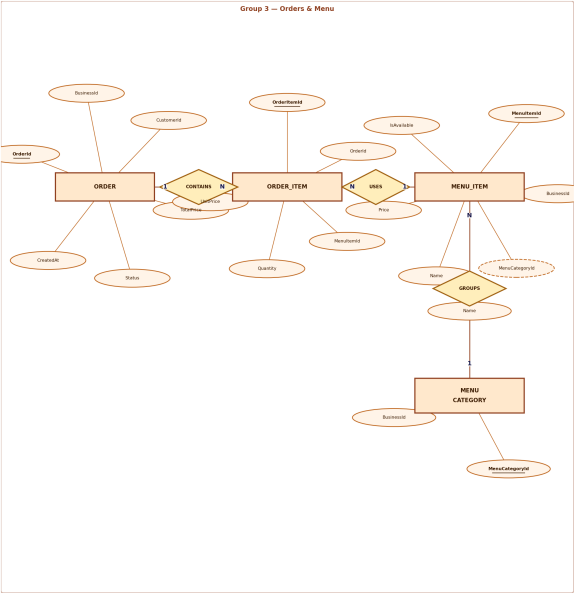
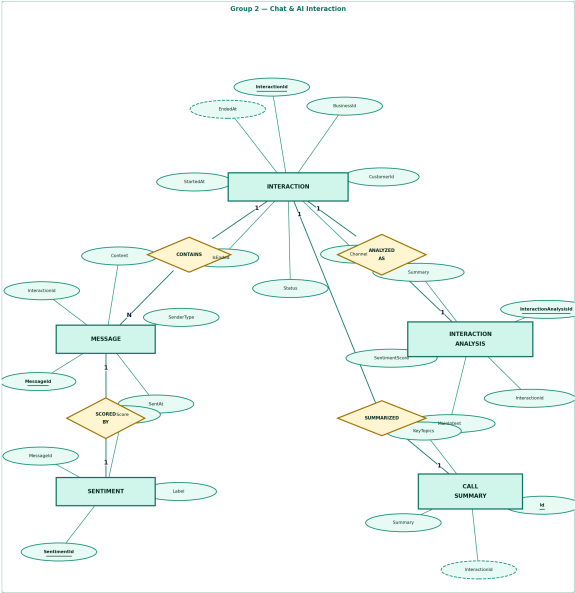
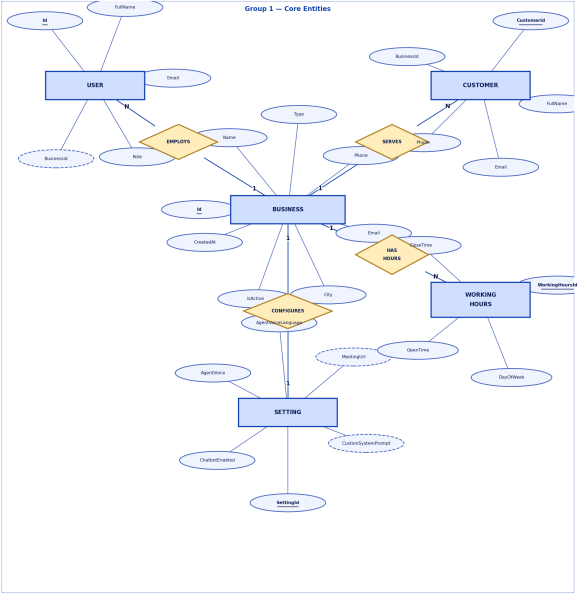


Chen ERD — Business Management Domain

Chen ERD — All Groups Combined (5-panel view)

Entity-Relationship Diagram (Traditional Chen Notation) — Digital Employee Platform

Rectangles = Entities | Ellipses = Attributes (underlined = PK, dashed = optional) | Diamonds = Relationships | 1/N = Cardinality



Chen ERD — All Groups Combined (5-panel view)

Chen ERD — Full System (All 23 Entities)

Rectangles = Entities | Ellipses = Attributes (underlined = PK, dashed = optional) | Diamonds = Relationships | 1/N = Cardinality



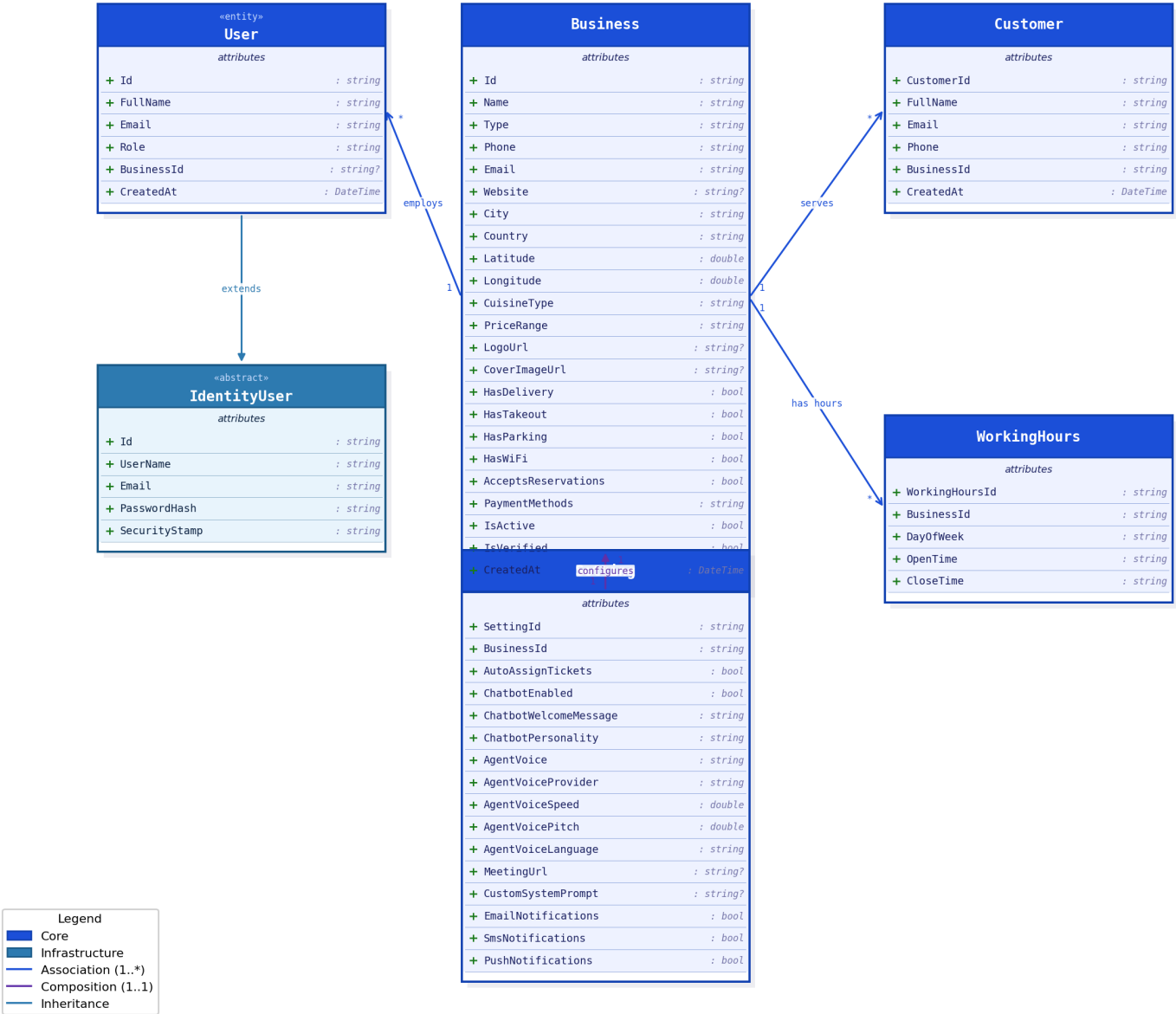
Section 3 — UML Class Diagrams

UML class diagrams showing entity classes with visibility (+/-), typed attributes, and method signatures. Inheritance (■) and association arrows connect related classes.

Class Diagram — Core Domain

Class Diagram — Group 1: Core Domain Classes

Business is the multi-tenant root. User (extends IdentityUser) and Customer are scoped per Business.

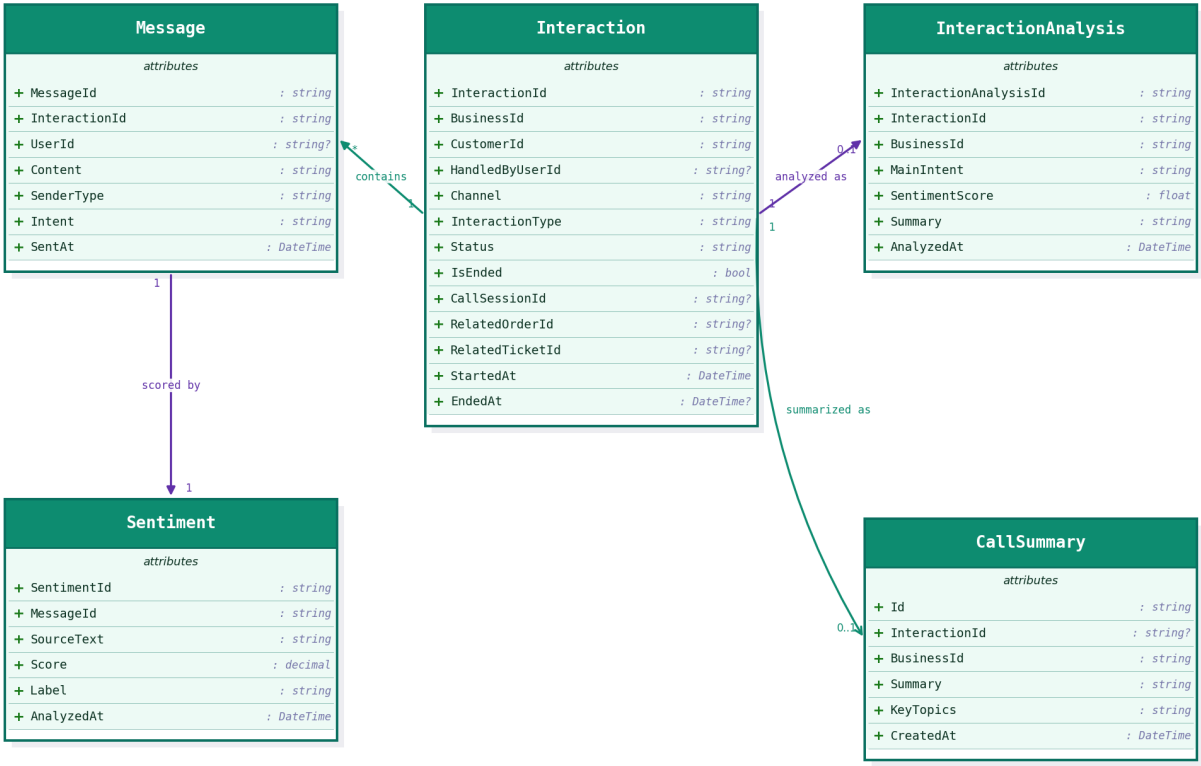


Class Diagram — Core Domain

Class Diagram — Chat & Interaction Domain

Class Diagram — Group 2: Chat & AI Interaction Classes

Tracks every conversation (channel-agnostic). Each Message may be scored by Sentiment; each Interaction may be summarised by AI.



Legend

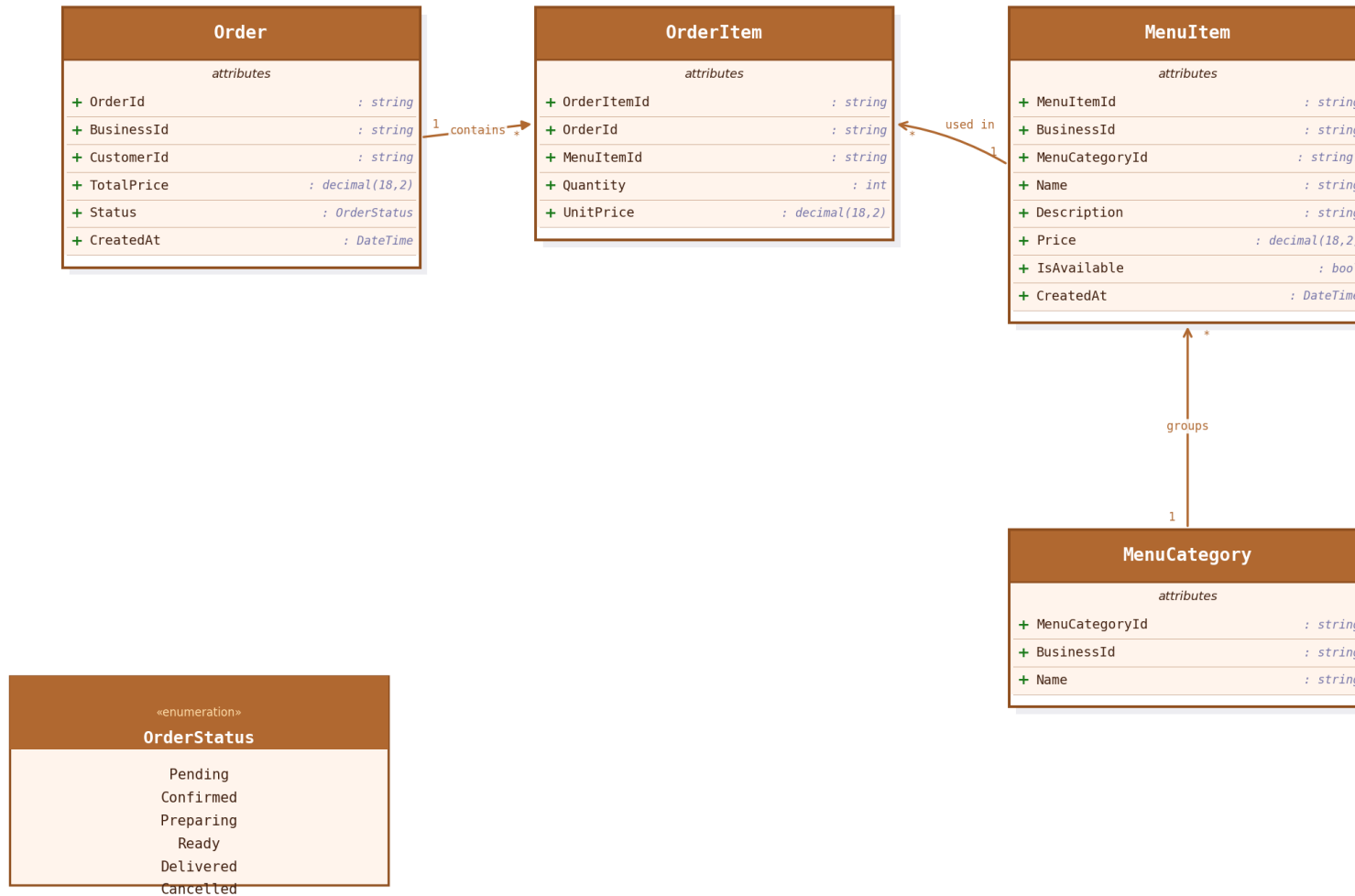
- Chat/AI
- Association / Composition
- Composition (1..1 cascade)

Class Diagram — Chat & Interaction Domain

Class Diagram — Orders & Menu Domain

Class Diagram — Group 3: Orders & Menu Classes

Restaurant menu catalogue, customer orders and line-items. Prices stored with decimal(18,2) precision.

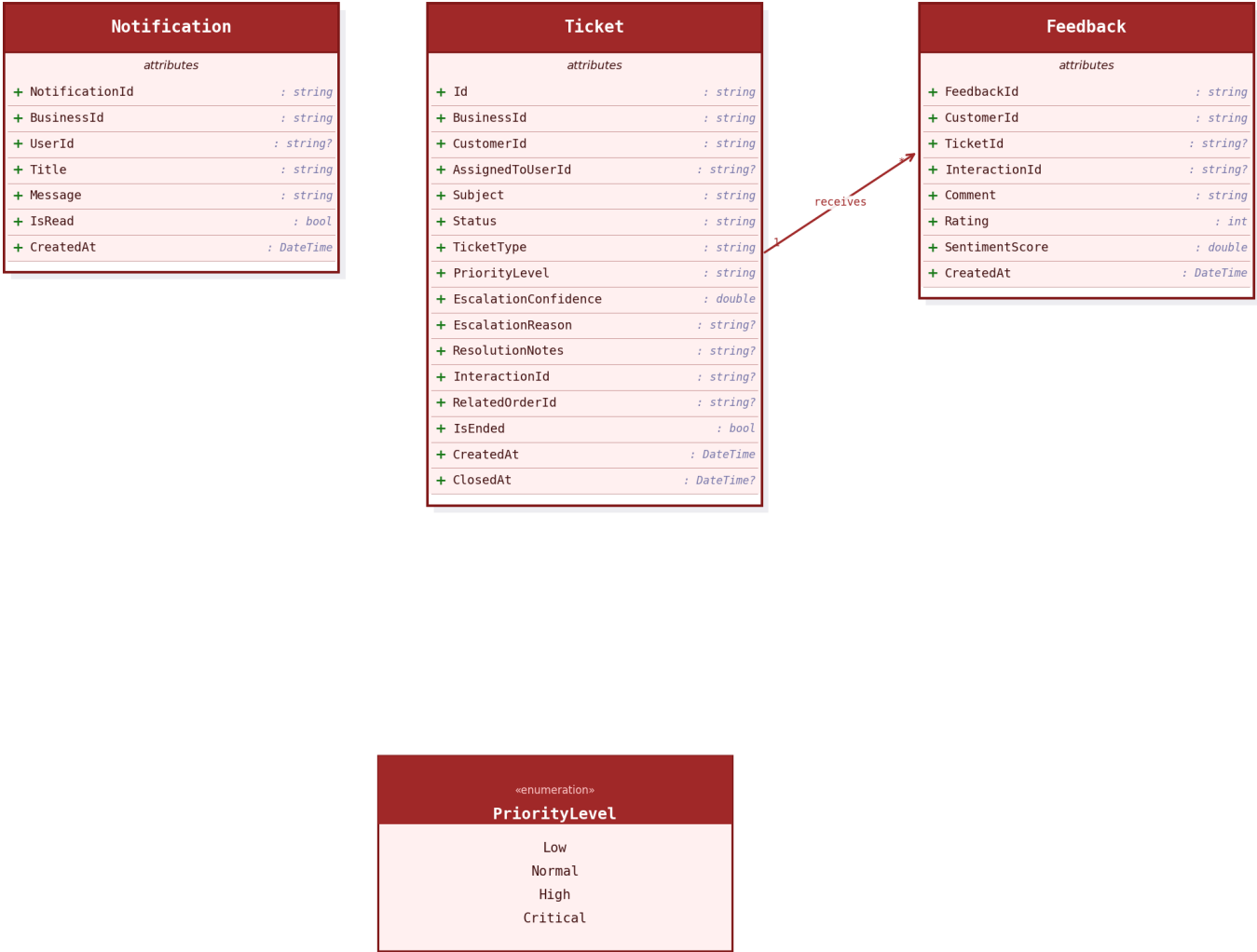


Class Diagram — Orders & Menu Domain

Class Diagram — Support & Notifications Domain

Class Diagram — Group 4: Support Classes

Support tickets raised by customers, collected feedback (with AI sentiment), and user push notifications.

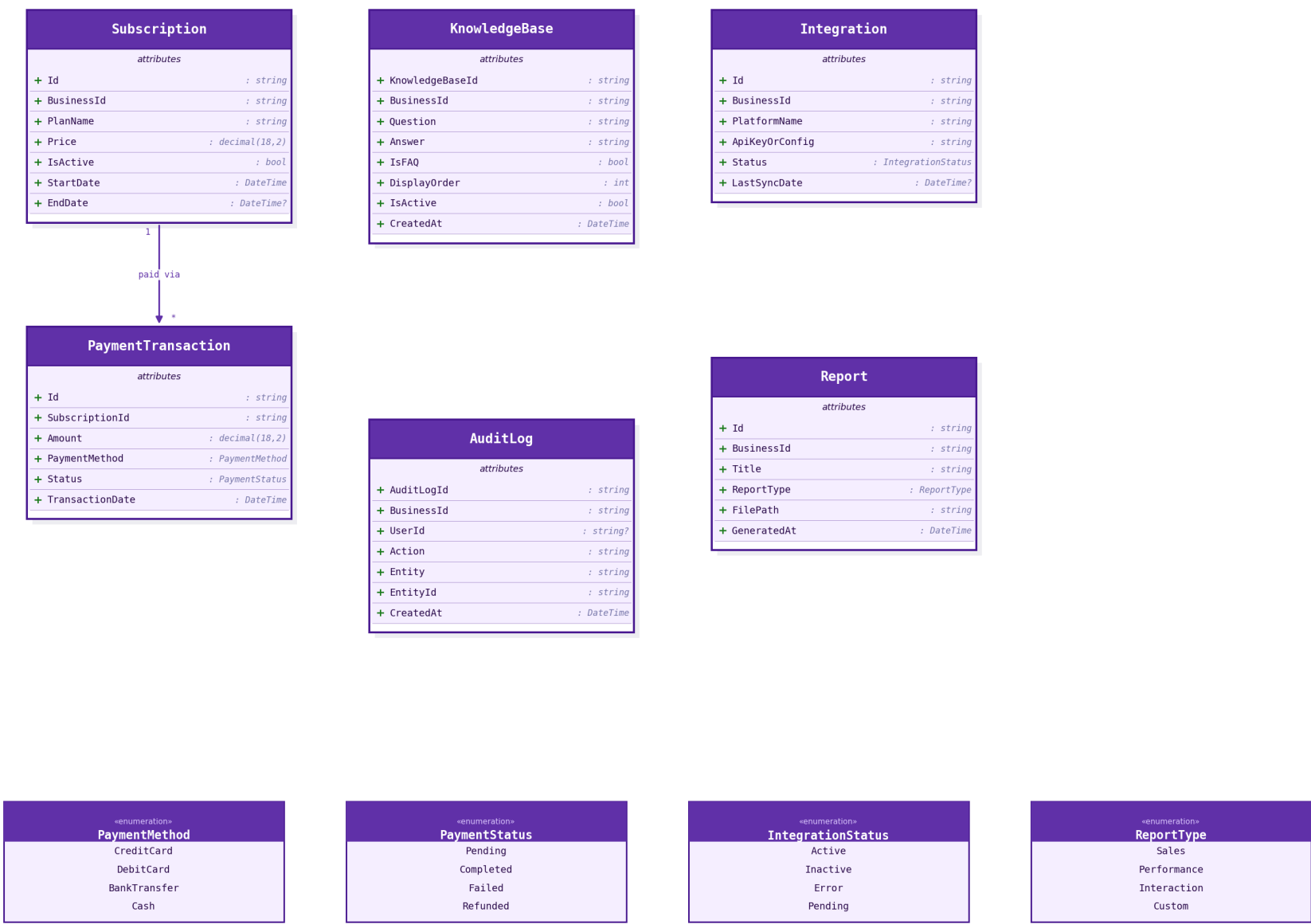


Class Diagram — Support & Notifications Domain

Class Diagram — Business Management Domain

Class Diagram — Group 5: Business Management Classes

Subscriptions, payments, AI knowledge base, third-party integrations, audit trail, and auto-generated reports.

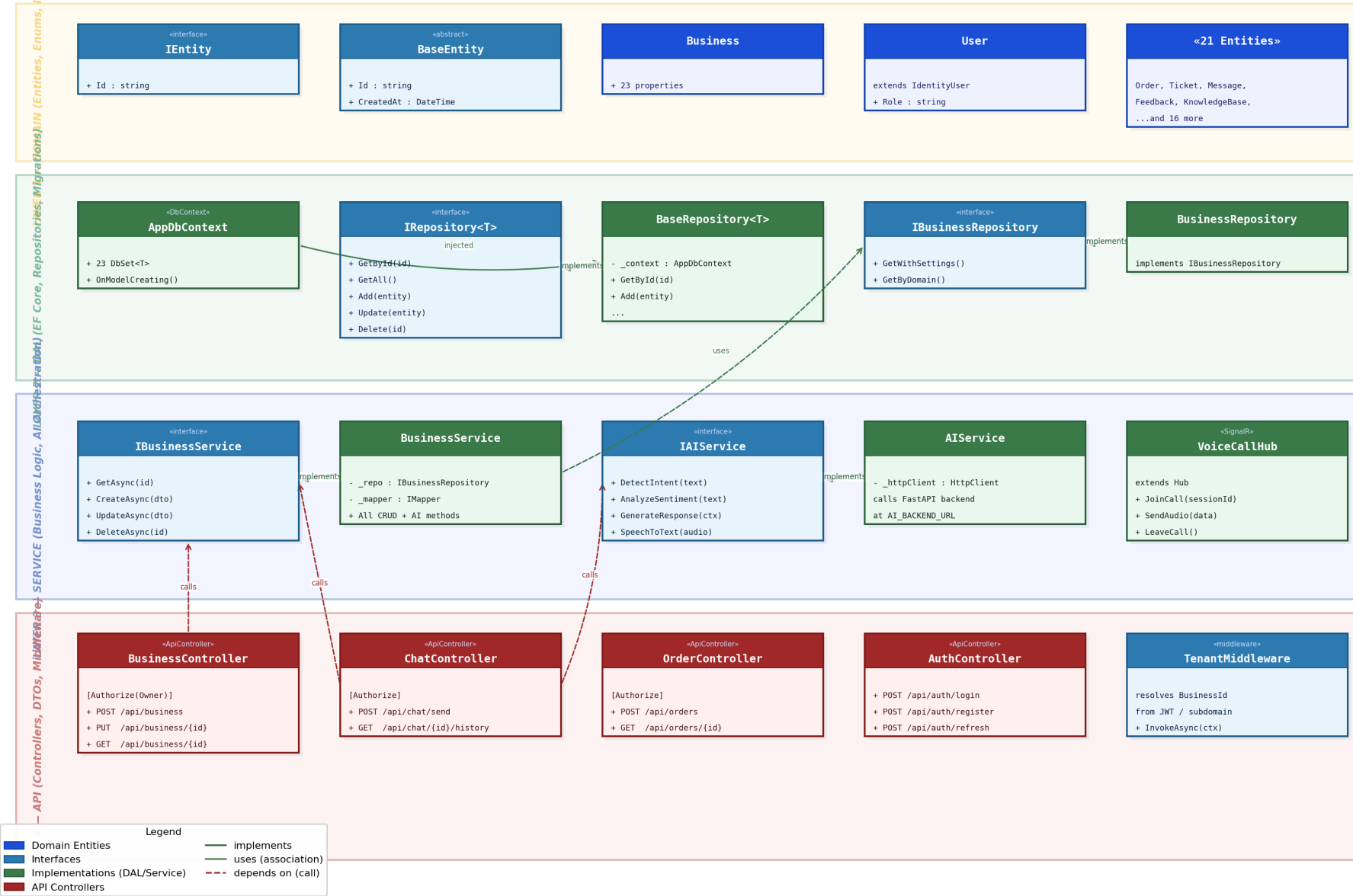


Class Diagram — Business Management Domain

Class Diagram — Architecture Overview (Clean Architecture layers)

Class Diagram — Clean Architecture Overview

Four-layer ASP.NET Core 9 architecture: Domain → DAL → Service → API. Each layer depends only on inner layers.



Class Diagram — Architecture Overview (Clean Architecture layers)

Class Diagram — Full System Combined

UML Class Diagrams — Digital Employee Backend

Clean Architecture - ASP.NET Core 9 - Multi-Tenant - 23 Domain Entities

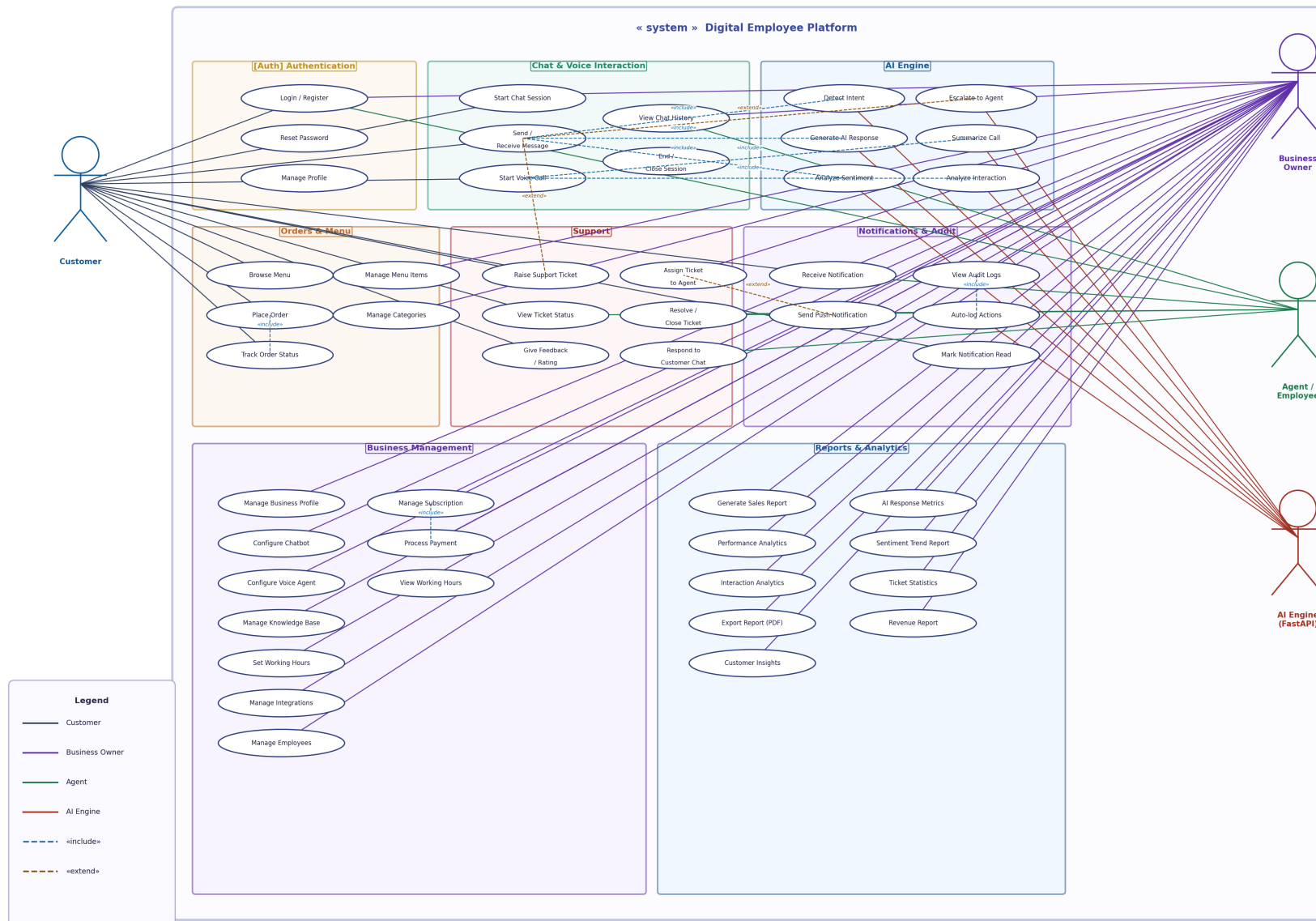


Section 4 — UML Use Case Diagram

UML Use Case diagram with 4 actors (Customer, Staff, Admin, AI Agent), 8 subsystems, and ~50 use cases. «include» and «extend» dependencies shown with dashed arrows.

Use Case Diagram — Digital Employee Platform

Multi-Tenant AI-Powered Customer Service · ASP.NET Core 9



UML Use Case Diagram — Full System